CALCULUS · UNIT 4: APPLICATIONS OF DERIVATIVES

Package Design Studio

Use optimization, related rates, and curve analysis to design packaging for a juice company.

CP2

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Fresh Squeeze Juice is launching a new drink and needs packaging that uses as little material as possible. The bottling plant fills cone-shaped tanks, and the marketing team wants to understand the sales curve.

You will use derivatives to design the package, control the filling process, and analyze sales.



Can Volume

355 cm³

(12 fl oz)

Box Sheet

30 cm by 20 cm

cardboard

Learning Goals

- I can find critical points and use them to find maximums and minimums.
- I can use related rates with a given formula.
- I can describe where a function increases and decreases.

Project Timeline

When	What you will do
Day 1	Open Box
Day 2	Filling the Tank
Day 3	Sales Curve
Final	Design Proposal + reflection

Part 1: Open Box

Day 1 · Cut squares of side x from each corner of a 30 cm by 20 cm sheet and fold up the sides.

Design Data Cards

Open Box	Can	Tank
Cut squares of side x	$V = 355$ cm³	Inverted cone
from a 30 × 20 sheet	minimize surface area	height 6 m, radius 2 m

Fill Rate	Sales	Cost
2 m³ per minute	$S(t) = -t^3 + 12t^2 + 60t$	Aluminum 0.02¢ per cm²
	thousand cans, t months	

1. Write the volume $V(x) = x(30 - 2x)(20 - 2x)$.

   Starter: *Multiply out $x(30 - 2x)(20 - 2x)$*

2. What is the domain of x?

3. Find $V'(x)$ and the critical point in the domain.

4. Find the maximum volume.

Part 2: Filling the Tank

Day 2 · Juice fills a tank at 2 m^3 per minute.

5. A cylinder tank has radius 2 m, so V = 4πh. Find $\frac{dh}{dt}$.

6. Explain why $\frac{dh}{dt}$ is constant for a cylinder.

7. A cone tank fills faster at the bottom. Explain why using its shape.

8. If juice drains at 0.5 m^3/min while filling at 2 m^3/min, what is the net $\frac{dV}{dt}$?

Part 3: Sales Curve

Day 3 · $S(t) = -t^3 + 12t^2 + 60t$ (thousand cans).

9. Find $S'(t)$ and when sales are increasing on $0 \le t \le 12$.

10. When are total sales at a maximum? Find the value.

11. Write one recommendation for the company using your results.

Final Product: Design Proposal

Your final product must include:

- Optimal box with justification
- Tank rate calculations
- Sales analysis
- One recommendation
- Design proposal

Reflection

R1. Why do you check the endpoints and critical points?

R2. What does $S'(t) = 0$ mean for sales?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Optimization, related rates, and curve analysis are set up, solved, and justified correctly. Goes beyond the minimum with a creative, realistic choice.	Optimization, related rates, and curve analysis are set up, solved, and justified correctly.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Proposal includes diagrams, derivative work, justifications, and recommendations. Polished, organized, and visually engaging.	Proposal includes diagrams, derivative work, justifications, and recommendations. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.